

SYS 3060 - Project Proposal Form

Team Members:

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Background Information and Motivation:

Last week and this week, the weather in Charlottesville has been highly inconsistent, with rapid and unexpected changes occurring over very short periods of time. For example, temperatures exceeded 80°F on Wednesday, followed by snow and hail the very next day, along with a tornado threat earlier this week. These abrupt transitions highlight the unpredictable and dynamic nature of Virginia weather patterns. Weather systems are inherently stochastic and influenced by a wide range of interacting atmospheric factors, making them difficult to predict with certainty. One common approach to modeling such systems is through a Discrete Time Markov Chain, where time is divided into discrete intervals and each interval is associated with a specific weather state, such as “hot,” “cold,” or “rainy.” This framework allows for the probabilistic modeling of transitions between different weather conditions over time.

Problem Statement:

The unpredictable weather changes observed in Charlottesville in recent weeks raise questions about the underlying transition behavior between different weather states. This system will be modeled using a Discrete Time Markov Chain to assess the frequency and likelihood of sudden weather changes. By estimating the transition probabilities between different weather states, this project seeks to better understand the conditions under which abrupt changes occur. Additionally, the analysis will examine whether any sequences between states can influence or

trigger an abrupt change in the weather. Insights from this project can reveal the mechanisms behind abrupt weather transitions and improve short-term forecasting approaches.

Project Scope, Boundaries, and Expected Assumptions:

This project will only look at weather in the Charlottesville local area. We limited our project in this scope because if we expanded to a bigger area it would be harder to classify the weather at a time interval into one general state. In addition, this project will look at past daily weather in the area to calculate empirical transition probabilities and determine weather patterns. We will analyze data in NOAA's Climate Data Online's Daily Summaries for Charlottesville zip code 22904 from 2010 to 3/18/2026 to better reflect current weather. Our model assumes that transitions between weather states are Markovian. Weather states are also classified into discrete states based on a combination of temperature binning and weather types available in the dataset to extend beyond general "hot", "cold", or "cloudy" states for weather modeling.

Goal of the Project:

The goal of our project is to develop a Discrete Time Markov Chain (DTMC) model that captures the probabilistic transition between the different states of weather in the Charlottesville area. Using historical weather data, we aim to estimate the transition probabilities between each of the states that we will define and analyze the likelihood of abrupt weather changes between each day.

This model will reveal patterns and sequences of weather states that are more likely to precede sudden transitions. Understanding these patterns is important because it allows us to highlight scenarios in which significant shifts in weather are more likely. In addition, we plan on

evaluating the effectiveness of the DTMC in short-term weather prediction by comparing the predicted state transitions with actual observed weather data.

Since weather patterns are often looked at more than one day in advance, we propose to improve our base model by extending it to predict multiple days into the future. With this multi-step forecasting model we can compare the prediction accuracy of the next-day model and the long-term predictive behavior. Overall, the objective of this project is to better understand the stochastic behavior of the local weather patterns and show how probabilistic modeling can be used to provide insight to short-term weather variability.

Expected Deliverables:

- Presentation Deck about the problem, how the model works, and the insights found by creating our model
- Project write-up in SIEDS paper template
- Code, datasets, simulation files and any other relevant project documentation

Additional Comments (not required):